

1 Methodology

We evaluate three portfolio construction models:

$$m \in \{\text{Triple Sort, AP-Trees, AP-Trees + Residual Momentum}\}.$$

For each model, let

$$x_{t+1}^{(m)} \in \mathbb{R}^{K_m}$$

denote the vector of candidate portfolio returns from month t to $t + 1$, where K_m is the number of candidate portfolios generated by model m . Let

$$w_t^{(m)} \in \mathbb{R}^{K_m}$$

denote the optimizer weights assigned to candidate portfolios at time t .

1.1 Model 1: Triple Sort

The Triple Sort model forms candidate portfolios by sorting stocks each month on three characteristics:

$$\text{LME, OP, Investment.}$$

Stocks are assigned to characteristic buckets, and each bucket portfolio is value-weighted using market equity:

$$w_{i,p,t}^{\text{stock}} = \frac{\text{ME}_{i,t}}{\sum_{j \in p_t} \text{ME}_{j,t}},$$

where p_t is the set of stocks in portfolio p at month t .

The candidate portfolio return is:

$$x_{p,t+1} = \sum_{i \in p_t} w_{i,p,t}^{\text{stock}} r_{i,t+1}.$$

1.2 Model 2: AP-Trees

The AP-tree model recursively partitions the cross-section using the fixed characteristic set:

$$\mathcal{C} = \{\text{LME, OP, Investment}\}.$$

We use tree depth:

$$D = 4,$$

and binary splits:

$$q = 2.$$

For the full AP-tree candidate set, we generate all feature sequences:

$$\mathcal{S} = \mathcal{C}^D,$$

so the number of feature-order trees is:

$$|\mathcal{S}| = 3^4 = 81.$$

Each node portfolio is value-weighted:

$$w_{i,p,t}^{\text{stock}} = \frac{\text{ME}_{i,t}}{\sum_{j \in p_t} \text{ME}_{j,t}},$$

and candidate portfolio returns are:

$$x_{p,t+1} = \sum_{i \in p_t} w_{i,p,t}^{\text{stock}} r_{i,t+1}.$$

1.3 Model 3: AP-Trees with Residual Momentum Tilt

For the residual momentum extension, we keep the same AP-tree structure but tilt stock weights within each node using a residual momentum signal.

We estimate residual momentum using:

$$\text{MOM_LOOKBACK} = 12, \quad \text{MOM_SKIP_RECENT} = 1, \quad \text{BETA_WINDOW} = 36.$$

Within each AP-tree node p , let $z_{i,t}$ be the cross-sectionally standardized residual momentum signal. The tilted stock weight is:

$$\tilde{w}_{i,p,t}^{\text{stock}} = \frac{w_{i,p,t}^{\text{base}} \exp(\tau z_{i,t})}{\sum_{j \in p_t} w_{j,p,t}^{\text{base}} \exp(\tau z_{j,t})},$$

where

$$\tau = 0.50.$$

The residual-momentum-tilted candidate return is:

$$\tilde{x}_{p,t+1} = \sum_{i \in p_t} \tilde{w}_{i,p,t}^{\text{stock}} r_{i,t+1}.$$

2 Optimization Variants

For each model m , we evaluate four variants.

2.1 A1: Static Paper-Style Optimizer, No Transaction Costs

This is the paper-faithful baseline. We use a fixed train/validation/test split with:

$$N_{\text{train+valid}} = 360, \quad CV_N = 3.$$

We estimate moments once using the first 360 months:

$$\hat{\mu}^{(m)} = \frac{1}{N_{\text{train+valid}}} \sum_{t=1}^{N_{\text{train+valid}}} x_t^{(m)},$$

$$\hat{\Sigma}^{(m)} = \text{Cov} \left(x_1^{(m)}, \dots, x_{N_{\text{train+valid}}}^{(m)} \right).$$

We solve:

$$\min_w \frac{1}{2} w^\top \hat{\Sigma}^{(m)} w + \lambda_1 \|w\|_1 + \frac{1}{2} \lambda_2 \|w\|_2^2$$

subject to:

$$w^\top \mathbf{1} = 1,$$

and, when imposed,

$$w^\top \hat{\mu}^{(m)} \geq \mu_0.$$

Our parameters are:

$$\lambda_1 = 0, \quad \lambda_2 = 10^{-3}, \quad \mu_0 = \text{None}, \quad w_i \geq 0.$$

The out-of-sample gross return is:

$$R_{t+1}^{\text{gross},(m)} = (w^{(m)})^\top x_{t+1}^{(m)}.$$

For the pure paper-style no-TC baseline:

$$R_{t+1}^{\text{net},(m)} = R_{t+1}^{\text{gross},(m)}.$$

2.2 A2: Static Optimizer with Stock-Level Transaction Costs

This variant keeps the static optimizer weights fixed, but applies transaction costs based on changes in the underlying stock holdings.

Final stock weights are:

$$W_{i,t}^{\text{stock},(m)} = \sum_{p=1}^{K_m} w_p^{(m)} w_{i,p,t}^{\text{stock},(m)}.$$

Stock-level turnover is:

$$TO_t^{\text{stock},(m)} = \sum_i \left| W_{i,t}^{\text{stock},(m)} - W_{i,t-1}^{\text{stock},(m)} \right|.$$

Transaction cost is:

$$\text{Cost}_t^{(m)} = c \cdot TO_t^{\text{stock},(m)},$$

where

$$c = 0.0025.$$

Net return is:

$$R_{t+1}^{\text{net},(m)} = R_{t+1}^{\text{gross},(m)} - \text{Cost}_{t+1}^{(m)}.$$

2.3 B: Rolling TC-Aware Optimizer with Portfolio-Level Turnover

For the rolling optimizer, we estimate moments using a rolling window:

$$L = 120.$$

At each month t , we estimate:

$$\hat{\mu}_t^{(m)} = \frac{1}{L} \sum_{s=t-L}^{t-1} x_{s+1}^{(m)},$$

$$\hat{\Sigma}_t^{(m)} = \frac{1}{L} \sum_{s=t-L}^{t-1} \left(x_{s+1}^{(m)} - \hat{\mu}_t^{(m)} \right) \left(x_{s+1}^{(m)} - \hat{\mu}_t^{(m)} \right)^\top.$$

We solve:

$$\min_w \frac{1}{2} w^\top \hat{\Sigma}_t^{(m)} w + \lambda_1 \|w\|_1 + \frac{1}{2} \lambda_2 \|w\|_2^2 + \lambda_{tc} \|w - w_{t-1}^{(m)}\|_1$$

subject to:

$$w^\top \mathbf{1} = 1,$$

and, when imposed,

$$w^\top \hat{\mu}_t^{(m)} \geq \mu_0.$$

Parameters are:

$$\lambda_1 = 0, \quad \lambda_2 = 10^{-3}, \quad \lambda_{tc} = 0.0025, \quad \mu_0 = \text{None}, \quad w_i \geq 0.$$

Portfolio-level turnover is:

$$TO_t^{\text{port},(m)} = \left\| w_t^{(m)} - w_{t-1}^{(m)} \right\|_1.$$

Transaction cost is:

$$Cost_t^{(m)} = c \cdot TO_t^{\text{port},(m)}, \quad c = 0.0025.$$

Gross and net returns are:

$$R_{t+1}^{\text{gross},(m)} = (w_t^{(m)})^\top x_{t+1}^{(m)},$$

$$R_{t+1}^{\text{net},(m)} = R_{t+1}^{\text{gross},(m)} - Cost_{t+1}^{(m)}.$$

2.4 C: Rolling TC-Aware Optimizer with Stock-Level Turnover

This is the most realistic implementation. The optimizer is the same as in Variant B, but transaction costs are computed using final stock-level holdings.

The final stock weight is:

$$W_{i,t}^{\text{stock},(m)} = \sum_{p=1}^{K_m} w_{p,t}^{(m)} w_{i,p,t}^{\text{stock},(m)}.$$

Stock-level turnover is:

$$TO_t^{\text{stock},(m)} = \sum_i \left| W_{i,t}^{\text{stock},(m)} - W_{i,t-1}^{\text{stock},(m)} \right|.$$

Transaction cost is:

$$Cost_t^{(m)} = c \cdot TO_t^{\text{stock},(m)}, \quad c = 0.0025.$$

Net return is:

$$R_{t+1}^{\text{net},(m)} = R_{t+1}^{\text{gross},(m)} - Cost_{t+1}^{(m)}.$$

3 Performance Metrics

For each model and variant, we compute monthly gross and net returns:

$$R_t^{\text{gross}}, \quad R_t^{\text{net}}.$$

Mean returns:

$$\bar{R}^{\text{gross}} = \frac{1}{T} \sum_{t=1}^T R_t^{\text{gross}}, \quad \bar{R}^{\text{net}} = \frac{1}{T} \sum_{t=1}^T R_t^{\text{net}}.$$

Annualized mean returns:

$$\mu_{\text{ann}}^{\text{gross}} = 12\bar{R}^{\text{gross}}, \quad \mu_{\text{ann}}^{\text{net}} = 12\bar{R}^{\text{net}}.$$

Annualized volatility:

$$\sigma_{\text{ann}}^{\text{gross}} = \sqrt{12}\sigma(R_t^{\text{gross}}), \quad \sigma_{\text{ann}}^{\text{net}} = \sqrt{12}\sigma(R_t^{\text{net}}).$$

Annualized Sharpe ratios:

$$SR_{\text{ann}}^{\text{gross}} = \sqrt{12} \frac{\bar{R}^{\text{gross}}}{\sigma(R_t^{\text{gross}})}, \quad SR_{\text{ann}}^{\text{net}} = \sqrt{12} \frac{\bar{R}^{\text{net}}}{\sigma(R_t^{\text{net}})}.$$

Sharpe decay due to transaction costs:

$$\Delta SR = SR_{\text{ann}}^{\text{gross}} - SR_{\text{ann}}^{\text{net}}.$$

Cumulative wealth:

$$W_t^{\text{gross}} = \prod_{s=1}^t (1 + R_s^{\text{gross}}), \quad W_t^{\text{net}} = \prod_{s=1}^t (1 + R_s^{\text{net}}).$$

Drawdown:

$$Peak_t = \max_{s \leq t} W_s^{\text{net}},$$

$$DD_t = \frac{W_t^{\text{net}}}{Peak_t} - 1.$$

Maximum drawdown:

$$MDD = \min_t DD_t.$$

Hit rates:

$$HitRate^{\text{gross}} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{R_t^{\text{gross}} > 0\},$$

$$HitRate^{\text{net}} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{R_t^{\text{net}} > 0\}.$$

Average turnover:

$$\overline{TO} = \frac{1}{T} \sum_{t=1}^T TO_t.$$

Average cost:

$$\overline{Cost} = \frac{1}{T} \sum_{t=1}^T Cost_t.$$

4 Summary of Parameter Values

Parameter	Value
Characteristics	{LME, OP, Investment}
AP-tree depth	4
Number of split buckets	2
Full AP-tree sequences	$3^4 = 81$
Residual momentum lookback	12
Residual momentum skip	1
Market beta window	36
Residual momentum tilt strength τ	0.50
$N_{\text{train+valid}}$	360
CV_N	3
Rolling optimizer window L	120
λ_1	0
λ_2	10^{-3}
λ_{tc}	0.0025
Transaction cost c	0.0025
μ_0	None
Long-only constraint	$w_i \geq 0$
Budget constraint	$w^\top \mathbf{1} = 1$

5 Plots

We generate the following diagnostic plots for each model family:

$$m \in \{\text{Triple Sort, AP-Trees, AP-Trees + Residual Momentum}\}.$$

Each method corresponds to a fully specified portfolio construction pipeline (including weighting scheme, optimizer, and transaction-cost specification). In practice, each plotted line corresponds to a distinct method label in the backtest.

We also include the S&P 500, proxied by SPY adjusted close, as a passive benchmark:

$$b = \text{S\&P 500 (SPY adjusted close)}.$$

5.1 Plot A: Cumulative Gross Returns

For each method m , we compute cumulative gross wealth:

$$W_t^{\text{gross},m} = \prod_{s=1}^t (1 + R_s^{\text{gross},m}). \quad (1)$$

For the benchmark:

$$W_t^{\text{SPY}} = \prod_{s=1}^t (1 + R_s^{\text{SPY}}). \quad (2)$$

This plot compares pre-transaction-cost performance across all strategies and against the passive market.

5.2 Plot B: Cumulative Net Returns

For each method m , cumulative net wealth is:

$$W_t^{\text{net},m} = \prod_{s=1}^t (1 + R_s^{\text{net},m}). \quad (3)$$

Net returns incorporate transaction costs:

$$R_t^{\text{net},m} = R_t^{\text{gross},m} - c \cdot TO_t^m, \quad (4)$$

where

$$c = 0.0025. \quad (5)$$

For no-transaction-cost variants:

$$R_t^{\text{net},m} = R_t^{\text{gross},m}. \quad (6)$$

For the S&P 500 benchmark:

$$W_t^{\text{net},\text{SPY}} = W_t^{\text{SPY}}, \quad (7)$$

since no transaction costs are applied.

5.3 Plot C: Gross vs Net Returns

For each method m , we plot gross and net wealth together:

$$W_t^{\text{gross},m} = \prod_{s=1}^t (1 + R_s^{\text{gross},m}), \quad (8)$$

$$W_t^{\text{net},m} = \prod_{s=1}^t (1 + R_s^{\text{net},m}). \quad (9)$$

Each method is plotted separately to visualize the impact of transaction costs on that specific strategy.

For the S&P 500:

$$W_t^{\text{gross},\text{SPY}} = W_t^{\text{net},\text{SPY}}. \quad (10)$$

5.4 Plot D: Net Drawdown

Define the running peak:

$$\text{Peak}_t^m = \max_{s \leq t} W_s^{\text{net},m}. \quad (11)$$

Drawdown is:

$$DD_t^m = \frac{W_t^{\text{net},m}}{\text{Peak}_t^m} - 1. \quad (12)$$

For the S&P 500:

$$DD_t^{\text{SPY}} = \frac{W_t^{\text{SPY}}}{\max_{s \leq t} W_s^{\text{SPY}}} - 1. \quad (13)$$

Maximum drawdown:

$$MDD^m = \min_t DD_t^m. \quad (14)$$

5.5 Plot E: Turnover

Turnover is computed at the implementable portfolio level:

$$TO_t^m = \sum_i |w_{i,t}^m - w_{i,t-1}^m|. \quad (15)$$

We plot the 12-month moving average:

$$\overline{TO}_t^m = \frac{1}{12} \sum_{s=t-11}^t TO_s^m. \quad (16)$$

For the S&P 500 benchmark:

$$TO_t^{\text{SPY}} = 0. \quad (17)$$

5.6 Plot F: Summary Metrics

For each method m , we compute:

$$SR_{\text{ann}}^{\text{gross},m} = \sqrt{12} \frac{\bar{R}^{\text{gross},m}}{\sigma(R^{\text{gross},m})}, \quad (18)$$

$$SR_{\text{ann}}^{\text{net},m} = \sqrt{12} \frac{\bar{R}^{\text{net},m}}{\sigma(R^{\text{net},m})}, \quad (19)$$

$$\overline{TO}^m = \frac{1}{T} \sum_{t=1}^T TO_t^m, \quad (20)$$

$$MDD^m = \min_t DD_t^m. \quad (21)$$

For the S&P 500:

$$\overline{TO}^{\text{SPY}} = 0. \quad (22)$$

We also report Sharpe decay due to transaction costs:

$$\Delta SR^m = SR_{\text{ann}}^{\text{gross},m} - SR_{\text{ann}}^{\text{net},m}. \quad (23)$$

For no-transaction-cost strategies and the benchmark:

$$\Delta SR = 0. \quad (24)$$